

Basics of Deep Learning

ICPBL Project Outline



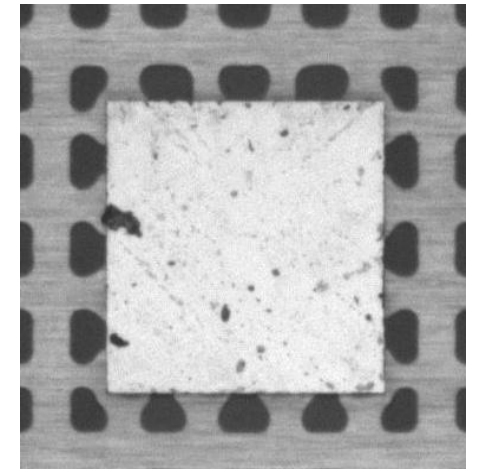
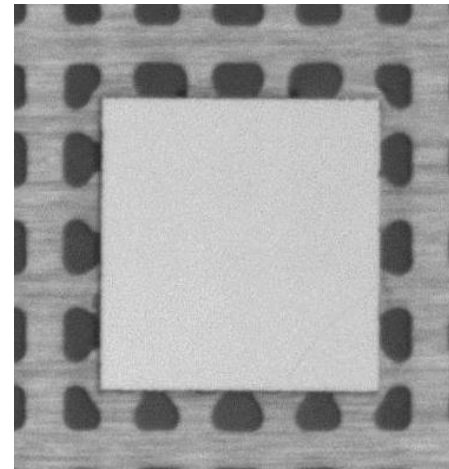
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MOTIVATION



- Wafer inspection is a key step for quality control in semiconductor manufacturing.
- Microscopic defects on wafers can reduce manufacturing yield.
- Defect pattern analysis is also needed for process improvement.
- We aims to build an AI-based semiconductor manufacturing inspection system for smart factories.



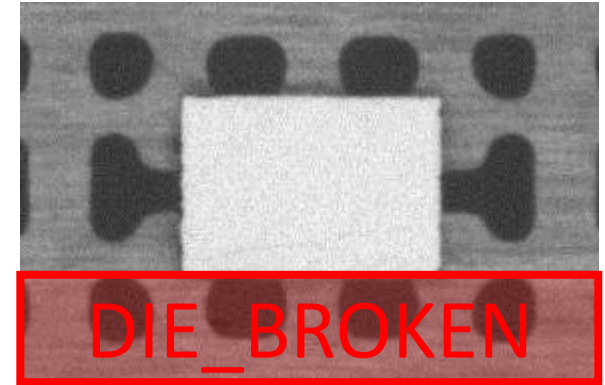
SURROUNDING CONDITIONS



- New defect types may appear in real manufacturing processes.
- A classifier trained only on existing defect classes may fail to handle unseen defect types.
- We aim to develop an open-set classifier that recognizes known classes and rejects unseen defects as “**Unknown**” class.

GOAL

- Given semiconductor wafer data containing various defect types, develop a deep learning model that can classify the defect condition of each sample.
 - Open-set classification task



IMPORTANT INFORMATION



- **Deadline:** 9th June 2026 6.00pm
- **Submit 2 files to LMS**
 1. **A pdf report**
 2. **A zip file containing your code**
 1. **notebook.ipynb:** a notebook file for your classification model
 - Please include open-set approach
 - Please include evaluation results and wandb visualization as well
 2. **model.pth:** your trained model weights

HOW TO PROCEED



1. Download wafer image dataset
2. Build a multi-class classifier model with **3 outputs**.
3. Train the baseline model using only the training data except unknown classes.
4. Validate you model on the validation set.
5. Evaluate your model and weights on test split (w/ unknown classes).
6. Improve your model to address open-set classification problem.

DATA PREPARATION

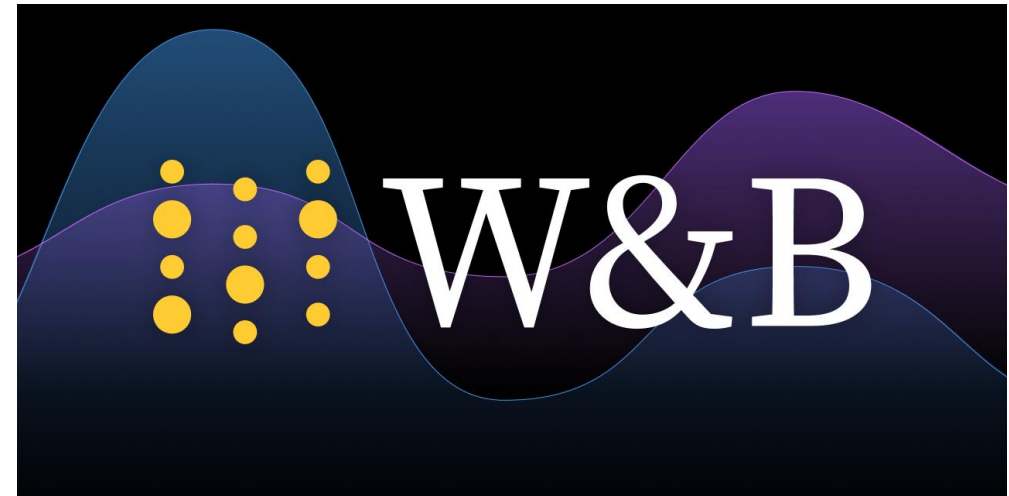
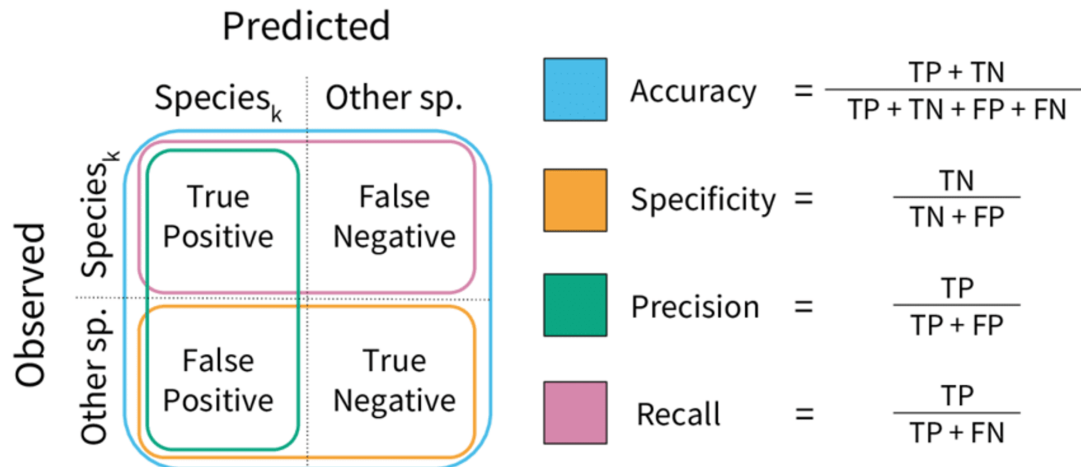


- Download [Data.zip](#) [download](#)
- The folder is divided into training (train) and validation (val) set.
- Each class folder contains the corresponding wafer images.
- The train set contains known classes (DIE_BROKEN, NORMAL, NO_DIE).
- The val/eval set contains known and unknown classes
- Each wafer folder includes five class folders (DIE_INK, NO_DIE, NORMAL, DIE_BROKEN, and DIE_CRACK).
- **Class labels should be assigned as follows.**
 - DIE_BROKEN: class 0
 - NORMAL : class 1
 - NO_DIE : class 2
 - DIE_CRACK, DIE_INK : “unknown” class

VISUALIZATION



- Visualize wafer dataset for each class
- Show training loss and training accuracy using WandB
- Show validation loss and validation accuracy using WandB
 - Accuracy: how many times you get things correct.
- Show precision, recall, and f1-score using WandB



FOR YOUR REPORT (MAX 6 PAGES)



- Data Preparation (5 points)
- Baseline model building (5 points)
- Training (20 points)
- Visualization (20 points)
 - Visualize wafer dataset
 - Plot loss, acc, precision, recall, f1-score per each epoch
- Discussions (30 points)
 - Improve your model (Open-set classification, Changing model architecture, etc.)
 - Ablation study on your modifications
 - Also discuss limitations of the current model
- Evaluation on the test set (10 points)
- Please submit the code and weights along the report PDF file.
- [NeurIPS format](#), max 6 page (excluding references)

IMPORTANT REMINDERS



- You're allowed to discuss with us or friends but please don't just copy each other's code or report.
 - **PLAGIARISM WILL NOT BE TOLERATED**
 - This exercise is designed to help you understand the training process.
 - When you get stuck in some errors whilst coding, the quickest way to try is search for a similar post on Google. (it's a good practice)

TA CONTACT POINTS



- **If you have any questions, please use the Q&A board on the LMS.**
- Jongwoo Lee (sanngu5@hanyang.ac.kr)
- Yurim Cho (csacyr@hanyang.ac.kr)
- Junyeong Yun (yuns0713@hanyang.ac.kr)



Your question or feedback is always welcome

Too fast or too slow? Too difficult or too easy?

Send message via e-mail or Zoom

